

THE STRUCTURE AND FUNCTION OF FERRITIN

PAULINE M. HARRISON

Department of Biochemistry
University of Sheffield
Sheffield S10 2TN, UK

Introduction During the past decade interest in metal metabolism has been stimulated. Research on the metabolism of iron, the major trace metal, has had its own momentum. Advances in our knowledge and understanding of an important part of this field, the structure and function of the iron-storage and transport proteins, ferritin and transferrin, have greatly accelerated. A review on transferrin recently appeared in this journal.¹ Here I summarise parallel developments on ferritin. A number of recent reviews which also give further references may be consulted for more comprehensive treatment of this subject.²⁻⁶

Ferritin, as its name implies, contains ferric iron, and its major and certainly best established physiological function is to store iron. This in itself gives it two roles. It provides a reserve of the element, which can be drawn upon as the need arises for the synthesis of molecules such as haemoglobin, cytochromes and iron-sulphur compounds. It also provides a means of safe-guarding the cell from a potentially toxic element. The toxicity of free iron is thought to arise from its ability to catalyse the activation of dioxygen leading to more reactive oxy radicals. Iron within ferritin is relatively inert. Hence it is not surprising that ferritin is ubiquitous within the plant and animal kingdoms and that its biosynthesis is stimulated by iron. Even bacteria have ferritin-like molecules, although these seem to have evolved quite separately, since there is no apparent amino acid sequence homology between ferritins and bacterioferritin⁷ (although data on the latter are limited). Curiously the iron of a second storage form, haemosiderin,⁸ which seems to be a breakdown product of ferritin and is to be found within secondary lysosomes or 'siderosomes', is said to be less accessible than that of ferritin. If this is correct a secondary protective mechanism may result from the rise in the relative proportion of haemosiderin to ferritin with increased iron loading. It may be of interest to note that the name 'ferritin' is derived from the Latin for iron 'ferrum', whereas 'haemosiderin' implies an origin in haemoglobin, not the presence of iron as haem. Haemoglobin is catabolized within the reticuloendothelial system, where its iron is temporarily stored.

In this review I shall discuss some of the properties of ferritin and what is known about the structures of its two components, the 'iron-core' and 'apoferritin', the protein shell which surrounds the iron-core. I shall then try to relate these structures to biological function(s) and give a brief account of the bacterioferritins.

General properties One of the distinguishing features of ferritin iron is its solubility, in contrast to the insolubility of haemosiderin. Ferric iron at physiological pH has a very low solubility of about 10^{-18} M. It tends to give polynuclear hydrolyses. These form small (<10 nm) particles which aggregate and precipitate. Encapsulated within a protein shell, however, they remain soluble. Ferritin's protein shell encapsulates up to 4500 Fe^{3+} atoms (see below for mechanisms), but cells contain a mixture of molecules of varying iron content, usually averaging not more than 3000 and often less. The protein shell contains 24 subunits of molecular weight each about 20 000. When the iron complex is added to the protein's 480 000 Da, molecular weights of up to nearly double this figure are attained (in 'full' ferritin). Iron-cores have a density about 2.5 times that of apoferritin, so the molecular volume allotted to them is only one quarter of the whole. Ferritin sediments in the analytical ultracentrifuge with a broad diffuse, pigmented, boundary (sedimentation coefficient averaging about 60S in horse spleen ferritin) and a sharp, 17.6S, colourless, boundary indicating the presence of some iron-free protein. The red-brown colour of ferritin is an aid to its purification. This colour disappears and the iron is released from the protein as Fe^{2+} by the action of reductants, such as sodium dithionite or thioglycolic acid. The iron-complex was first shown to be inside the protein by electron microscopy and this has been confirmed by X-ray and neutron scattering from solutions and X-ray diffraction of single crystals.

Iron-cores Chemically ferritin's iron-cores consist of a ferric-oxyhydroxy-phosphate complex.⁵ A similar complex (but lacking phosphate) is formed when solutions of ferric salts are warmed or brought gradually to neutral pH. The product is known nowadays as 'ferrihydrite', since it seems to be identical with a so-named poorly crystalline material found on the walls of disused mineworkings or in cold or hot springs where it may be produced with the aid of iron-oxidizing bacteria. Ferrihydrite, although related in structure to that of haematite (α -Fe₂O₃) differs from this mineral and from the α -, β -, γ - and δ -polymorphs of FeOOH. All are based, however, on combinations of FeO₆ octahedra, sharing corners, edges or faces (thus reducing the ratio Fe:O to less than 1:6). Ferritin iron-cores give several X-ray or electron diffraction lines indicating crystallinity and some of those seen by high resolution electron microscopy are single crystals about 6–7 nm across.⁹ Other cores contain two or three smaller crystals or show disordered regions. The crystallinity of the cores seems to be inversely related to their phosphate content. The rate of iron to phosphorus is quite variable ranging from very high values (phosphorus sometimes undetected) in some iron-loaded ferritins to below 2:1 in some bacterioferritins. About 9:1 is normal for horse spleen ferritin. Although probably attached to the protein's inner surface, at the atomic level no structural relationship between mineral and protein is evident.

Chemical properties of apoferritin *Amino acid composition* Apoferritins characteristically have a preponderance of acidic over basic residues (pIs usually between 4.5 and 5.5), a relatively high content of apolar groups and an absence of disulphide bridges (as expected for intracellular proteins), although cysteine may be present.

Amino acid sequences Sequences have been determined either as protein or as cDNA, for ferritins from horse,¹⁰ rat,¹¹ tadpole¹² and human.¹³ These sequences clearly exhibit polymorphism within one species, see Table 1. In humans two basic types of subunit are evident, known as H and L, which have different sizes as seen by electrophoresis in polyacrylamide gels containing sodium dodecylsulphate. The H and L subunits were so-called because of their respective preponderance in heart and liver.¹⁴ Two sequences have been determined in human liver which exhibit only 55% identities in amino acid positions. This is in striking contrast to the 85% identities found by comparison of sequences designated L from horse or human spleen or liver and rat liver ferritins (see Table 1). The tadpole (embryonic) red cell cDNA sequence, which is the most primitive so far determined (closest to the primordial ferritin), exhibits 67% identity with human H and 59% with human L of Table 1. In humans and rats multiple gene copies corresponding to both H and L subunits have been found, but it is not certain whether they are copies of identical or slightly different genes within the general type.^{15,16} The small sequence differences observed for L subunits from different sources (of the same species) may represent polymorphism or methodological aberrations.

Examination of the known amino acid sequences shows certain regions are highly conserved in L subunits, but there is only one region of sequence containing more than four residues that is identical throughout all known sequences. This is from 122 to 128 inclusive (Glu 130 is also conserved throughout, and replacements at 129 and 131 are conservative). The significance of this becomes more obvious when the three-dimensional structure of apoferritin is examined.

Three dimensional structure of apoferritin Although the only three-dimensional structure to have been determined *de novo* is that of horse spleen apoferritin,⁵ it is clear from preliminary X-ray diffraction analyses that ferritins from horse and dog spleen and liver, rat, mouse and human liver and human spleen are essentially isomorphous in structure. Close inspection of the distribution of amino acids in relation to the three-dimensional structure of horse spleen apoferritin lends support to this conclusion. Some differences in detail are of course apparent, such as an insertion of eight residues in the rat liver sequence, as compared to the other L subunits, between residues 157 and 158. The following description of the three-dimensional structure can be taken to apply to all these mammalian ferritins.

The conformation of the apoferritin subunit consists of a 4-helix bundle of helices A (res 10–39), B (45–72), C (92–120) and D (124–155) and a short helix E (160–169) lying at an

Table 1 Amino acid sequences of some ferritins

HoS-L (protein)	Horse Spleen — L subunit (ref 10)	HuL-H (cDNA)	Human Liver — H subunit (N-terminal T T A S not shown) (ref 13)
RaL-L (cDNA)	Rat Liver — L subunit (ref 11)	TaR-H (cDNA)	Tadpole Red Cell — H subunit (ref 12)
HuL-L (cDNA)	Human Liver — L subunit (ref 13)	RaL-Li 8 residue insertion in RaL-L between 157 and 158	

	5				10				15				20				25				30				35				40												
HoS-L	S	S	Q	I	R	Q	N	Y	S	T	E	V	E	A	A	V	N	R	L	V	N	L	Y	L	R	A	S	Y	T	Y	L	S	L	G	F	Y	F	D	R	D	
RaL-L	T	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	H	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
HuL-L	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	S	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
HuL-H	T	—	—	V	—	—	—	—	H	Q	D	S	—	—	—	I	—	—	Q	I	—	—	E	—	Y	—	—	—	V	—	—	—	M	S	Y	—	—	—	—	—	—

Diagonal = Total number of residues in sequence.

Upper Right = % Identity between two sequences.

Lower Left = Number of overlapping residues between two sequences.

acute angle (about 60°) to the bundle, Fig 1. The twenty-four subunits of identical conformation pack in such a way that their axes (and those of the long helices A to D) lie within the shell. An unusual feature of the subunit conformation compared with other proteins having the 4-helix bundle motif,¹⁷ is the presence of a long loop, L, of nineteen residues connecting opposite ends of the bundle through C- and N-terminal ends respectively of helices B and C. Operation of 4-, 3- and 2- fold symmetry axes gives the '432' quaternary structure shown in Fig 2. Pairs of helix bundles lie antiparallel, related by 2-fold symmetry. The long loops, L, of these pairs come together to form a very short region of anti-parallel pleated sheet lying on the outside surface of the shell. Subunits are packed tightly to give a shell thickness of nearly 2.5 nm (outside diameter about 12.5 nm and inside about 7.8 nm). The numerous intersubunit interactions (salt bridges and H-bonds as well as apolar interactions) presumably account for the high thermostability of the molecule. Eighty per cent of residues buried within the shell, including those in intersubunit interfaces, are either conserved or replaced conservatively within the known sequences.¹⁸ It is not surprising therefore that artificial hybrids (eg of rat and horse L subunits) can be made. In the case of human H and L subunits *all* residues so far identified as making strong intersubunit bonds are identical in both sequences. These H and L subunits should readily be able to make copolymers as had been postulated.¹⁴ Rat liver apoferritin molecules comprising about one third H-chain are structurally isomorphous with horse spleen apoferritin, implying that both H and L chains have the same conformation. Residues on the outside surface of the shell are, as may be expected, less

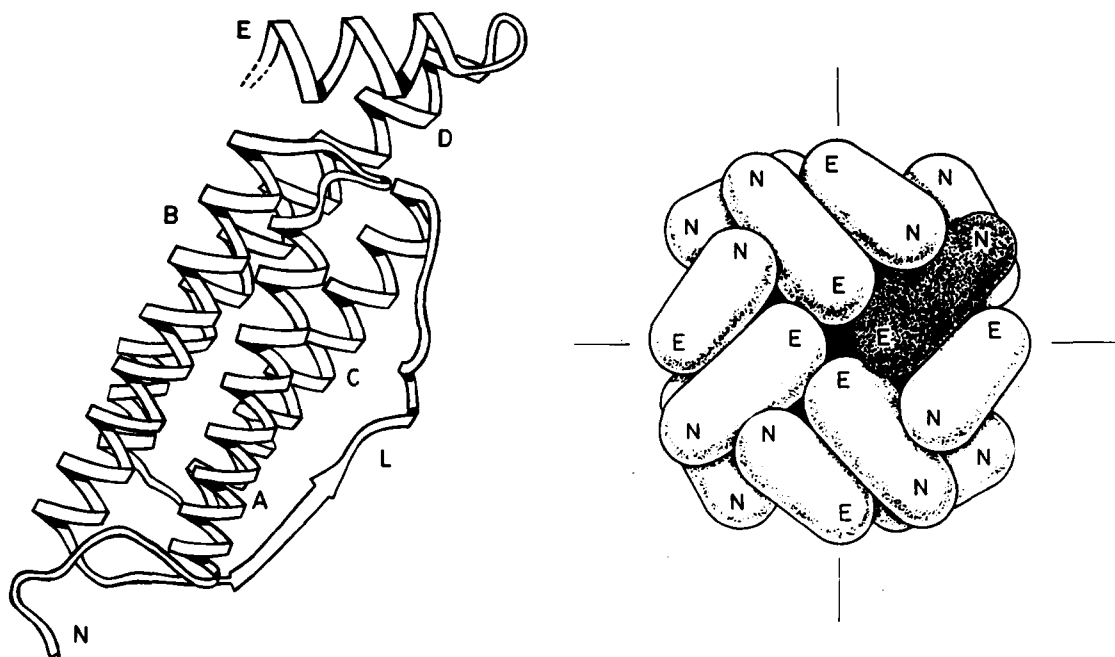


Figure 1 Schematic drawing of a ferritin subunit. The main features are the five helices, A to E, and the long inter-helix loop, L. The loop, L and the N-terminal residues, N, lie on the outside of the protein shell. Helix E runs from the outside to the inside, so that the C-terminal end of the chain is at the inside surface of the molecule

Figure 2 Schematic drawing of a ferritin molecule. Twenty-four subunits (sausages) surround the ferritin iron-core (black), which may contain up to 4500 Fe(III) as "ferrihydrite". N indicates the N-terminus and E the N-terminal end of the E helix (refer to Fig 1)

well conserved than those within the shell (at least two thirds of external positions allow substitutions). Conservation is no greater on the inside, cavity surface. This is rather surprising if there is contact between shell and core. However, such contact may be limited and tenuous, in view of their differences in atomic structure and lack of alignment.

Although there are tight intersubunit contacts within the shell, around the eight 3-fold and six 4-fold symmetry axes, there are gaps. In the next section I discuss how iron may be deposited inside the cavity and released from it. Transit pathways for iron and small molecules or ions mediating its uptake or release, may be through these gaps or channels. The region of structure bordering on a 3-fold axis is shown in Fig 3 and that around a 4-fold axis in Fig 4. The difference in character of these channels is immediately striking.⁵ Side chains near the 3-fold axes are predominantly hydrophilic with three aspartic (No 127) and three glutamic acid (No 130) residues pointing towards the axes. These residues are part of the conserved region noted above. Side chains immediately bordering the 4-fold axis channels are less well conserved and are either entirely or predominantly apolar. The central residue No 165, on the four E-helices surrounding these channels is leucine in all sequences, that near the outer exit being leucine or methionine and that nearest the cavity being leucine or histidine.

Although in the static crystal structure the average diameters of 3-fold and 4-fold axes are not more than about 0.4 nm (their lengths are respectively about 0.5 and 1.25 nm) it is clear from X-ray and neutron scattering measurements of apoferritin or ferritin solutions in the presence of sugars or other small molecules, from analysis of the kinetics of iron release and from X-ray crystallographic analysis of the position of covalently bound derivatives of apoferritin, that molecules larger than 0.4 nm (0.7 to 1.0 nm) can penetrate into the molecular interior.³ Hence ferritin, like other proteins, is not a rigid molecule of fixed structure, but undergoes conformational fluctuations sometimes referred to as 'breathing'.

Metal binding sites on apoferritin Several metal binding sites have been located by X-ray analysis in metal derivatives of horse spleen apoferritin,¹⁸ but I list in Table 2, only those which have one or more than one ligand that is conserved in all known sequences. These represent six of the eleven sites found in horse spleen apoferritin. On grounds of greatest ligand conservation, sites 1 to 4 are potentially the most likely to represent functional sites. The first three of these are in the 3-fold channels. It is tempting to hypothesize that these represent transit sites, the route by which iron enters the molecule (note that metals can occupy different positions and there is some ligand flexibility within these channels) and also that site 4 on the cavity surface may participate in ferrihydrite nucleation.

Alternatives are conceivable. Could ferrihydrite nucleation start in one of the 3-fold channels for example? Could hydrated Fe^{2+} enter through the 4-fold channels? Binding studies of Chasteen and colleagues¹⁹ by means of electron spin resonance measurements or with ion-selective electrodes show that a number of metal ions (Mn^{2+} , Cd^{2+} and VO^{2+}) bind initially with a stoichiometry of about 0.33 per subunit, which would place them in the 3-fold channels. It may be postulated that as further iron approaches these ions move into the cavity where hydrolysis and linkage into polymers occurs.

Table 2 Metal binding sites with conserved amino acid ligands

No	Ligands	Metals	Location
1	Asp127, Asp127' Asp127''	Cd^{2+} , Zn^{2+}	3-fold channel (inside)
2	Glu130, Glu130' Glu130'', 3H ₂ O	Cd^{2+} , Zn^{2+}	3-fold channel (outside)
3	Asp127, Asp127', Asp127'', Glu130, Glu130', Glu130''	Tb^{3+}	3-fold channel (middle)
4	Glu57, Glu60	Tb^{3+}	Cavity surface, on B helix
5	Glu57 (Glu136) H ₂ O	UO_2^{2+}	Cavity surface, between B and D helices
6	Glu63 (Arg59)	UO_2^{2+}	Cavity surface, between B and B'

Brackets denote residues not conserved in all known sequences. Other ligands are conserved. Prefixes ' and '' denote residues from neighbouring, symmetry related subunits.

Iron release processes Little is known about how ferritin yields its iron *in vivo*. *In vitro* mediators of release include reductants and small chelators.²⁻⁶ *In principle* these could act either by entering the molecule and interacting with the iron-core, gradually 'dissolving' it, or they could remove iron from sites on the protein's outer surface presumed to be in equilibrium with those on the iron-core. For reduction there is a third possibility, that electrons are tunnelled through the protein from reductant on its outside or embedded within it. With the exception of the large protein, transferrin, which is inefficient as a direct acceptor for ferritin iron (only about 5 Fe^{3+} /molecule/day are transferred in *in vitro* incubation experiments) most, if not all, of the chelators known to be capable of removing Fe^{3+} from ferritin almost certainly do so by penetration of the protein shell.³ These include citrate, nitrilotriacetate and desferrioxamine. The latter is a molecule of ~600 Da molecular weight used in the treatment of iron overload because of its very high affinity for ferric iron. The action of these molecules is quite slow, only some tens of Fe^{3+} atoms being released per molecule per hour.

Reductive mobilization can be much faster, although rapid release depends on the presence of molecules which bind Fe^{2+} with high affinity (2,2'-bipyridine which chelates through its nitrogen atoms is often used). It has recently been shown that in the absence of

an Fe^{2+} chelator, all the Fe^{3+} of the ferritin iron-core can be reduced to Fe^{2+} by stoichiometric amounts of sodium dithionite with *very little loss of Fe^{2+} from the molecule*.²⁰ The Fe^{3+} -core is apparently converted to an Fe^{2+} -core of unknown structure. The action of the Fe^{2+} -chelators may be two-fold. By competing for Fe^{2+} inside the molecule they may break up the Fe^{2+} -core. Secondly by binding the Fe^{2+} they may allow it to be released as a small complex in which its charge is shielded. This could facilitate its passage through the apolar 4-fold channels.

For the preparation of apoferritin from ferritin either dithionite or thioglycollate is generally used, reduction is facilitated and is usually carried out at low pH (4.6–5.1). A number of other thiol reagents have also been shown to mobilize ferritin iron, including cysteine and glutathione. These and ascorbic acid have been considered as possible physiological mediators of ferritin iron release, although the slow rates found *in vitro* shed doubt on their effectiveness. Reduced flavins (studied in the presence of 2,2'-bipyridine) allow ferritin iron to be mobilized much more rapidly. Their low cellular concentrations and their requirement for anaerobiosis call into question their physiological application for this purpose,²¹ although this has been proposed. In short, neither the means by which ferritin iron is removed within cells, nor the site of its removal is known. Iron cores must be lost when molecules are degraded but for what fraction of ferritin iron turnover this accounts remains to be determined.

Ferritin biosynthesis To exercise a protective function the biosynthesis of ferritin protein must be maintained such that a spare capacity for iron is always present (ie molecules on average must contain well under their full complement of Fe). The stimulation by iron of apoferritin's biosynthesis is the immediate cellular mechanism for dealing with this element.²² In fact ferritin can reach remarkably high levels increasing by 50-fold over normal. Studies with embryonic red cells indicate that mRNA levels are not raised by iron influx. Iron seems to act not by increasing the rate of transcription but by making ferritin mRNA more available for translation.^{4,23} It is uncertain how this is effected. Iron may alter the binding to ferritin mRNA of ferritin subunits, or of some other protein translational repressor, thus derepressing translation. Alternatively it may cause the mobilization of ferritin mRNA from mRNP particles. Once available it seems to be translated with high efficiency.^{4,23}

The induction of ferritin biosynthesis by iron may be very important in a number of pathological conditions. Iron release to the blood is decreased during inflammation and tissue ferritin rises. In thalassaemia treated by repeated blood transfusion, ferritin and haemosiderin levels increase to deal with the excess iron introduced and which cannot be eliminated through the kidneys.

Ferritin heterogeneity and its significance

Genetic and post-translational heterogeneity After much argument about the meaning of multiple bands in ferritin as observed by techniques such as isoelectric focussing and electrophoresis in denaturing gels, the heteropolymer hypothesis proposed to explain such phenomena, is now generally accepted. This hypothesis proposed by Drysdale and colleagues¹⁴ is that ferritin molecules are co-polymers of the two subunits, known as H and L. The proportions of H and L subunits vary across the isoelectric profile in a given tissue ferritin and also the relative proportions of H and L vary with tissue type.^{14,24} As mentioned above, the H subunit is usually predominant in ferritin from heart, (and at early stages of development and in some cancer tissue) and the L subunit in liver (and also spleen). In human liver or spleen H-rich ferritin fractions are more acidic than L-rich, and this is also true of rat liver ferritin, although the reverse is the case with ferritin from horse spleen. However, although such trends are present in other tissue ferritins (which may have different overall H:L levels), a mismatch between H:L content and iso-electric point has been observed when fractions of different tissue ferritins are compared.²⁴ Moreover subclasses of both subunit types have appeared when the subunits are separated by HPLC²⁵ or by two-dimensional gel electrophoresis.² It is not clear whether these are due to expression of different members of multiple L or H gene families in different tissues or arise from post-translational modifications.

There is evidence^{15,16} that ferritin H and L chains are derived from multigene families, and, in humans, that members of H and L gene families have several different chromosome

locations. Jain *et al*¹⁵ have found that there are about 12 H genes and about 8 L genes with a high degree of homology among members of each family. How many of these genes are expressed and how many represent pseudo-genes is uncertain. It is possible that different genes are preferentially expressed in different tissues and the finding of some amino acid sequence differences would support such a conclusion. However, some of the reported differences may be methodological artifacts.

There is also ample evidence that ferritin can be post-translationally modified. In some cases the modification has been manifested simply by a change in isoelectric point (following iron injection).²⁶ In others actual modification such as glycosylation,²⁷ phosphorylation,²⁸ proteolytic cleavage or inter-subunit cross-linking²⁹ have been identified. Thus, for example, much of the ferritin in human serum is glycosylated. Experimentally this glycosylation has been found to be associated with L chains.²⁷ Human L chains, and those of rat and horse, have a single *N*-glycosylation sequence Asn(7)-Tyr(8)-Ser(9), which is absent from human H-chain and from tadpole red cell ferritin sequences. This glycosylation sequence lies on the outside of the assembled molecule.¹⁸ Another such sequence [Asn(107)-Gln(108)-Ser(109)] is found in human H chains but not in the amphibian. However since Asn(107) occurs at an intersubunit interface, carbohydrate is probably not attached at this position.¹⁸ Hence the presence of non-glycosylated H-rich ferritins in serum may be an indicator that these have appeared through cell damage rather than secretion. H-rich and L-rich ferritins are distinguishable immunologically and preferential reactivity towards antiserum to heart-type (H-rich) ferritin has been noted in the serum ferritin of some cancer patients.

Physiological significance of isoferritins Isoferritin distribution patterns have been found to change during development or with pathological state but the significance of such changes is a matter for discussion. It has been suggested that the role of L-type ferritins is principally for iron-storage, whereas H-type ferritins are the more metabolically active, perhaps playing a role in intramolecular iron transport as well as in detoxification.³⁰ Thus in iron-rich tissue, a preponderance of L-rich ferritins is usual and may increase with iron-loading.²⁴ At the same time the amount of iron stored as haemosiderin increases both relatively and absolutely, so the synthesis of L subunits alone is unable to meet the demands for iron-storage. Evidence for the view that H-type ferritins are the more active in iron metabolism includes the observations that when human erythroid cells are incubated with ⁵⁹Fe-transferrin for 1 h more of the ⁵⁹Fe is found in H-type than in L-type ferritin, and also that, *in vitro*, higher rates of iron accumulation are associated with molecules of higher H content.³⁰ However, although one would naturally expect functional differences to be associated with H and L subunits, it has been shown that other factors such as molecular iron content³ or intersubunit cross-linking²⁹ may be able to override any influence of subunit composition on iron uptake rates measured *in vitro*. Several additional functions have been suggested for ferritins, the most controversial being that H-type ferritins have a specific role in the control of haemopoiesis through a negative feed-back mechanism.^{30,31} One proposal not specifically related to isoferritins is discussed in the next section.

Ferritin metal binding

This article has discussed our current understanding of ferritin's major and best established function as an iron detoxicant and store. In these functions it bears some resemblance to the metallothioneins although structurally these molecules could hardly be more different.³² In addition to binding zinc and copper, metallothionein binds the toxic element, cadmium, and indeed is induced by it. It is also known to bind and to be induced by several other metals including, Ag, Hg and Bi. We have mentioned that apoferritin is capable of binding several metals other than iron. This is also true of ferritin's iron-cores. These are found to bind at least 800 g atoms of Fe and they also bind Al, Zn and Cu. Hence ferritin may have a secondary role in the detoxification of other metals.³³

Bacterioferritin and ferritin evolution

Ferritin is widely distributed throughout the animal and plant kingdoms (it is found in vertebrates and invertebrates, flowering plants and fungi). Bacterial species also have their own type of ferritin. Thus there appears to be a universal need for a means of removing and storing iron, arising from its biological usefulness coupled with the toxicity

of free iron and the insolubility of the ferric species. The need for ferritin must have developed some 2.5 billion years ago in parallel with dioxygen production by photosynthetic organisms. Divergence of H and L subunits from a single primordial ferritin is thought to have occurred much more recently, eg about 300 million years ago.³⁴ Amphibia also appeared at about this time. The relatively close similarity of the amphibian amino acid sequence, Table 1, and its mRNA with those of both H and L mammalian subunits suggests that it may be close to that of the primordial precursor of mammalian ferritin.¹² Bacterial ferritin (from *Escherichia coli*) on the other hand is quite different in its amino acid sequence.⁷ The molecule also differs from ferritin in apparently having two functions, since it contains haem (one per two subunits) as well as non-haem iron.³⁵ Nevertheless its protein is a hollow shell composed of twenty-four crystallographic subunits arranged in the same symmetry as ferritin, and it has an iron-core. Thus here we have convergence of two different proteins to the same task. This further emphasises the importance of these fascinating proteins.

References

- ¹ Ching-Ming Chung, M (1984) *Biochem Educ* **12**, 146–154
- ² Aisen, P and Listowsky, I (1980) *Ann Rev Biochem* **49**, 357–393
- ³ Clegg, G A, Fitton, J E, Harrison P M and Treffry, A (1981) *Progr Biophys Molec Biol* **36**, 53–86
- ⁴ Theil, E C (1983) in *Adv Inorg Biochem* **5**, 1–38 (edited by Theil, E C, Eichhorn, G L and Marzilli, L G), Elsevier, New York
- ⁵ Ford, G C, Harrison, P M, Rice, D W, Smith, J M A, Treffry, A, White, J L and Yariv, J (1984) *Phil Trans Roy Soc B* **304**, 551–565
- ⁶ Crichton, R R (1984) *Trends Biochem Sci* **9**, 283–286
- ⁷ Tsugita, A and Yariv, J (1985) *Biochem J* **231**, 209–212
- ⁸ Selden, C, Owen, H, Hopkins, J M P and Peters, T J (1980) *Brit J Haematol* **44**, 593–603
- ⁹ Massover, W H and Cowley, J M (1973) *Proc Natl Acad Sci USA* **70**, 3847–3851
- ¹⁰ Heusterspreute, M and Crichton, R R (1981) *FEBS Lett* **129**, 322–327
- ¹¹ Leibold, E A, Aziz, N, Brown, A J P and Munro, H N (1984) *J Biol Chem* **259**, 4327–4334
- ¹² Didsbury, J R, Theil, E C, Kaufman, R E and Dickey, L F (1986) *J Biol Chem* **261**, 949–955
- ¹³ Boyd, D, Vecoli, C, Belcher, D M, Jain, S K and Drysdale, J W (1985) *J Biol Chem* **260**, 11755–11761
- ¹⁴ Arosio, P, Adelman, T G and Drysdale, J W (1978) *J Biol Chem* **253**, 4451–4458
- ¹⁵ Jain, S K, Barrett, K J, Boyd, D, Favreau, M F, Crampton, J and Drysdale, J W (1985) *J Biol Chem* **260**, 11762–11768
- ¹⁶ Brown, A J P, Leibold, E A and Munro, H N (1983) *Proc Natl Acad Sci USA* **80**, 1265–1269
- ¹⁷ Richardson, J S (1981) *Adv Prot Chem* **34**, 167–339
- ¹⁸ Harrison, P M, White, J L, Smith, J M A, Ford, G C, Rice, D W, Addison, J M and Treffry, A (1985) in *Proteins of Iron Storage and Transport* (edited by Spik, G, Montreuil, J, Crichton, R R and Mazurier, J), Elsevier Science Publishers, Amsterdam, pp 67–79
- ¹⁹ Wardeska, J G, Viglione, B and Chasteen, N D (1986) *J Biol Chem* **261**, 6677–6683
- ²⁰ Watt, G D, Frankel, R B and Papaefthymiou, G C (1985) *Proc Natl Acad Sci USA* **80**, 3640–3643
- ²¹ Crichton, R R (1985) in *Proteins of Iron Storage and Transport* (edited by Spik, G, Montreuil, J, Crichton, R R and Mazurier, J), Elsevier Science Publishers, Amsterdam, pp 99–110
- ²² Drysdale, J W and Munro, H N (1966) *J Biol Chem* **241**, 3630–3637
- ²³ Shull, G E and Theil, E C (1982) *J Biol Chem* **257**, 14187–14191
- ²⁴ Bomford, A, Conlan-Hollingshead, A and Munro, H N (1981) *J Biol Chem* **256**, 948–955
- ²⁵ Collawn, J F Jr, Lau, P Y, Morgan, S L, Fox, A and Fish, W W (1985) *Comp Biochem Physiol* **81B**, 901–904
- ²⁶ Treffry, A, Lee, P J and Harrison, P M (1984) *Biochem J* **220**, 717–722
- ²⁷ Cragg, S J, Wagstaff, M and Worwood, M (1981) *Biochem J* **199**, 565–571
- ²⁸ Ihara, K, Maeguchi, K, Young, C T and Theil, E C (1984) *J Biol Chem* **259**, 278–283
- ²⁹ Mertz, J R and Theil, E C (1983) *J Biol Chem* **258**, 11719–11726
- ³⁰ Jacobs, A, Hodgetts, J and Hoy, T G (1984) in *Ferritins and Isoferritins as Biochemical Markers* (edited by Albertini, A *et al*), Elsevier Science Publishers BV pp 113–127
- ³¹ Broxmeyer, H E, Bognacki, J, Dorner, M and de Souza, M (1981) *J Exp Med* **153**, 1426–1444
- ³² Kojima, Y and Kagi, J H R (1978) *Trend Biochem Sci* **90–92**
- ³³ Price, P J and Joshi, J G (1984) *Toxicology* **31**, 151–163
- ³⁴ Munro, H N, Leibold, E A, Vass, J K, Aziz, N, Rogers, J, Murray, M and White, K (1985) in *Proteins of Iron Storage and Transport* (edited by Spik, G, Montreuil, J, Crichton, R R and Mazurier, J), Elsevier Science Publishers BV Amsterdam, pp 331–341
- ³⁵ Yariv, J, Kalb, A J, Sperling, R, Bauminger, E R, Cohen, S G and Ofer, S (1981) *Biochem J* **197**, 171–175